Laplace Transform Networks

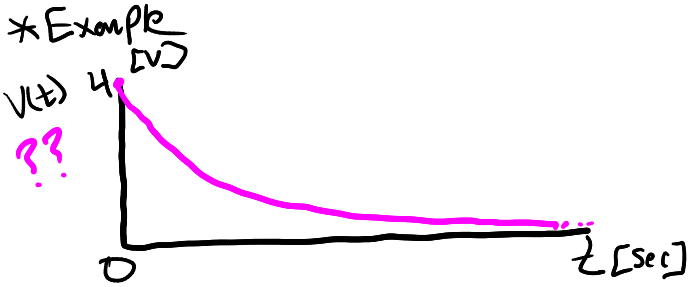
Unilateral $(0,\infty)$

Bilateral $(-\infty,\infty)$

Laplace Transform

$$\mathcal{L}[v(t)] = \int_{0^-}^{\infty} v(t)\, e^{-st}\, dt = V(s)$$

*Example



$$v(t) = \begin{cases} ?? & t < 0^- \\ 4e^{-9t} & t \geq 0^- \end{cases}$$

$$\mathcal{L}[v(t)] = \int_{0^-}^{\infty} 4e^{-9t} e^{-st}\, dt = V(s)$$

$$\Rightarrow 4\int_{0^-}^{\infty} e^{-(s+9)t}\, dt = 4 \times \frac{1}{-(s+9)} e^{-(s+9)t} \Big|_{t \to 0^-}^{t \to \infty}$$

$$= \frac{-4}{s+9}\left[\underbrace{e^{-(s+9)(+\infty)}}_{0} - \underbrace{e^{-(s+9)(0^-)}}_{1}\right]$$

$$= \frac{4}{s+9}$$

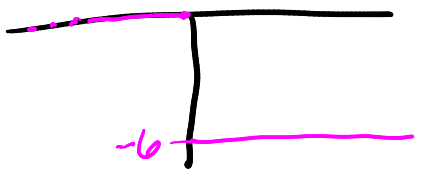
$$v(t) = 4e^{-9t} \xleftrightarrow{\mathcal{L}} \frac{4}{s+9} \quad \text{for } t \geq 0$$

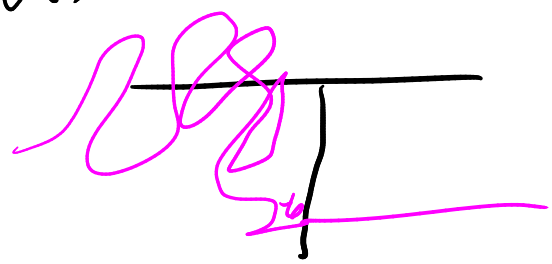
or

$$be^{at} \xleftrightarrow{\mathcal{L}} \frac{b}{s-a}$$

*Example

$v(t) = -6$ for $t > 0$ Find $V(s)$





$$\mathcal{L}[-6] = \mathcal{L}[-6u(t)] = V(s)$$

Let $b = -6$, $a = 0$

$$be^{at} \xleftrightarrow{\mathcal{L}} \frac{b}{s-a}$$

$$\Rightarrow \frac{-6}{s}$$

$$-6 \xleftrightarrow{\mathcal{L}} \frac{-6}{s}$$

$$-6u(t) \longleftrightarrow \frac{-6}{s}$$

$$f(t) \xleftrightarrow{\mathcal{L}} F(s) \Rightarrow g(t) \xleftrightarrow{\mathcal{L}} G(s)$$

Consider $g(t) = e^{at} f(t)$ $\quad G(s) = ?$

$$G(s) = \int_{0^-}^{\infty} g(t) e^{-st} dt = \int_{0^-}^{\infty} e^{at} f(t) e^{-st} dt$$

$$= \int_{0^-}^{\infty} f(t) e^{-(s-a)t} dt$$

Using the definition of Laplace transform

$$F(s) = \int_{0^-}^{\infty} f(t) e^{-st} dt$$

$$\Rightarrow G(s) = F(s-a)$$

$$g(t) = e^{at} f(t) \xleftrightarrow{\mathcal{L}} F(s-a)$$

* Example

$$i(t) = \cos(8t) \text{ for } t > 0$$

$$I(s) = \int_{0^-}^{\infty} \cos(8t) e^{-st} dt = \int_{0^-}^{\infty} \left(\frac{e^{j8t} + e^{-j8t}}{2} \right) e^{-st} dt$$

$$be^{at} \longleftrightarrow \frac{b}{s-a}$$

$b = 1$

$a = j8$

$$= \frac{1}{2} \left[\int_{0^-}^{\infty} (e^{j8t} + e^{-j8t}) e^{-st} dt \right]$$

$$= \frac{1}{2} \left[\int_{0^-}^{\infty} e^{j8t} e^{-st} dt + \int_{0^-}^{\infty} e^{-j8t} e^{-st} dt \right]$$

$$I(s) = \frac{1}{2} \left[\frac{1}{s-j8} + \frac{1}{s+j8} \right]$$

$$\Rightarrow \frac{1}{2} \left[\frac{(s+j8) + (s-j8)}{(s-j8)(s+j8)} \right] = \frac{1}{2} \left[\frac{2s}{s^2 + 8^2} \right]$$

$$\Rightarrow I(s) = \frac{s}{s^2 + 8^2} = \boxed{\frac{s}{s^2 + 64}}$$

$$\cos(ct) \xleftrightarrow{\mathcal{L}} \frac{s}{s^2 + c^2}$$

$g(t) = 7e^{-4t}\cos(3t)$, Let $f(t) = \cos(3t)$

$$F(s) = \frac{s}{s^2+3^2}$$

$$G(s) = 7F(s+4)$$

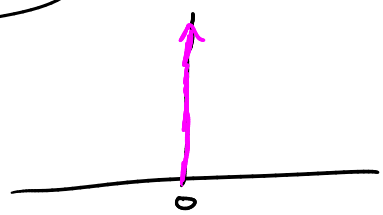
$$\Rightarrow G(s) = \frac{7(s+4)}{(s+4)^2+3^2} = \boxed{\frac{7(s+4)}{(s+4)^2+9}}$$

* For sine

$$\boxed{\sin(ct) \xleftrightarrow{\mathcal{L}} \frac{c}{s^2+c^2}}$$

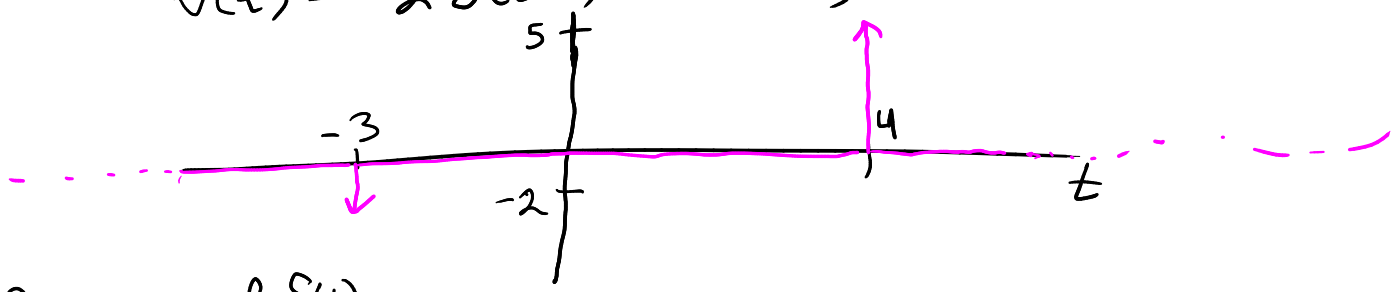
* Delta Function (Impulse, Dirac)

$$\delta(t) \equiv \frac{du(t)}{dt}$$



Area = 1

$$v(t) = -2\delta(t+3) + 5\delta(t-4)$$



* Properties of $\delta(t)$

Definition: $\delta(t) = 0$ for $t \neq 0$

$$\int_a^b \delta(t)\,dt = 1 \quad \text{if } t = 0 \text{ is within } [a,b]$$

Sifting property of $\delta(t)$

$$\int_{-\infty}^{\infty} \delta(t) f(t)\,dt = \int_{0^-}^{0^+} \delta(t) f(t)\,dt = f(0)$$

$$\int_{-\infty}^{\infty} \delta(t-9) f(t)\,dt = f(9)$$

$$\int_{-2}^{\infty} \delta(t+3) f(t)\,dt = 0$$

$$\mathcal{L}[\delta(t)] = \int_{0^-}^{\infty} \delta(t) e^{-st}\, dt = e^{-s(0)} = 1$$

$$\delta(t) \xleftrightarrow{\mathcal{L}} 1$$

$$u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s}$$

$$e^{-at} \longleftrightarrow \frac{1}{s+a}$$

Ex

$$v(t) = \delta(t) + e^{-4t} u(t) \quad \text{for } -\infty < t < \infty$$

$$V(s) = 1 + \frac{1}{s+4} = \boxed{\frac{s+5}{s+4}}$$ improper fraction

Suppose $I(s) = 7\,\dfrac{s+6}{s+2}$ $\quad i(t) = ?$ for $t \geq 0$

$$\begin{array}{r} 1 \\ s+2 \overline{\big) s+6} \\ -(s+2) \\ \hline 4 \end{array} \quad \Rightarrow \frac{s+6}{s+2} = 1 + \frac{4}{s+2}$$

$$I(s) = 7 + \frac{28}{s+2}$$

$$\boxed{i(t) = 7\delta(t) + 28e^{-2t}\ A \quad \text{for } t \geq 0}$$

*Ex

$$V = \frac{4s-36}{s^2-4s-21} = \frac{4(s-9)}{(s+3)(s-7)}$$ Partial Fraction Decomposition

$$\Rightarrow V(s) = \frac{A}{s+3} + \frac{B}{s-7}$$ Where the solution will take the form

$$v(t) = Ae^{-3t} + Be^{7t} \quad \text{for } t \geq 0$$

Use Heaviside Covering Method (Residue Method)

$$V(s) = \frac{4s-36}{(s+3)(s-7)} = \frac{A}{s+3} + \frac{B}{s-7}$$

A/

$$V(s)(s+3)\Big|_{s\to -3} = \underbrace{\frac{A}{\cancel{s+3}}\cancel{(s+3)}\Big|_{s\to -3}}_{A} + \underbrace{\frac{B}{s-7}\cancel{(s+3)}\Big|_{s\to -3}}_{0}$$

$$A = \frac{4s-36}{\cancel{(s+3)}(s-7)}\cancel{(s+3)}\Big/_{s\to -3} = \frac{4(-3)-36}{(-3-7)} = 4.8$$

$$B = \frac{4s-36}{(s+3)\cancel{(s-7)}}\cancel{(s-7)}\Big/_{s\to 7} = \frac{4(7)-36}{(7+3)} = -0.8$$

$$V(s) = \frac{4.8}{s+3} + \frac{-0.8}{s-7}$$

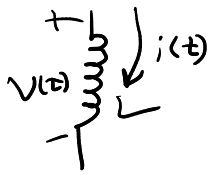
$$\boxed{v(t) = 4.8e^{-3t} - 800\times10^{-3}e^{7t} \text{ V for } t \geq 0}$$

*Laplace Transform Networks

$$f(t) \xleftrightarrow{\mathcal{L}} F(s)$$

$$\mathcal{L}\left[\frac{df(t)}{dt}\right] = sF(s) - f(0)$$

Remember,



$$v(t) = L\frac{di}{dt}$$

$$i(t) = C\frac{dv}{dt}$$

* take $i(t) = C\frac{dv}{dt}$

$$\mathcal{L}[i(t)] = I(s) = C[sV(s) - v(0)]$$

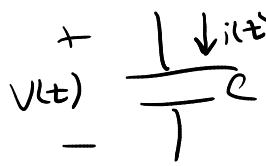
$$I = \underbrace{Cs}_{\text{Siemens}}V - CV(0) \quad [\text{Amp}\cdot\text{sec}]$$

$$\Rightarrow I = I_1 + I_2$$

* Suggests parallel typology

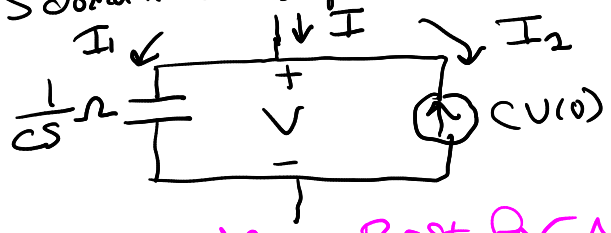
Therefore

Time Domain



$\Rightarrow$

s domain or Laplace Domain



This does NOT create any new nodes... Best for Nodal Analysis

Worst for Mesh Analysis

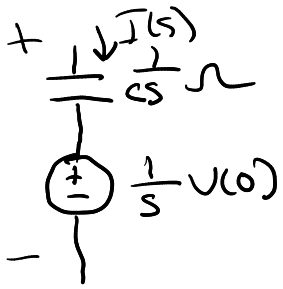
Consider the other case

$$I = CsV - CV(0)$$

$$\Rightarrow V = \underbrace{\frac{1}{Cs}}_{\Omega}\underbrace{I}_{A} + \frac{1}{s}V(0) \Rightarrow V = V_1 + V_2$$

Series typology

$\frac{1}{T} \Rightarrow$



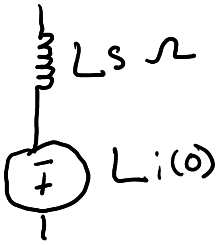
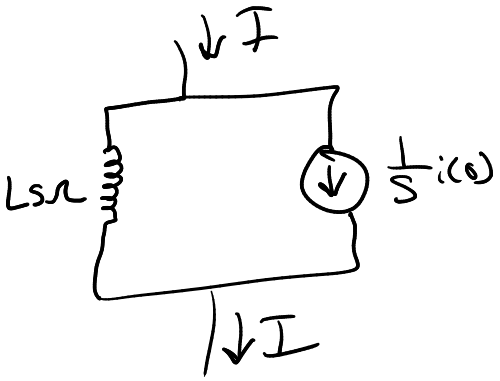
Best for Mesh

Worst for Nodal

For inductor

$V(t) = L\frac{di}{dt}$ $\quad \mathcal{L}[V(t)] = \underbrace{LsI}_{V} - Li(0)$

$\Rightarrow$



$I = \frac{1}{Ls}V + \frac{1}{s}i(0) \Rightarrow$